Executive Summary

Top 3 Defect Types:

Structural defects – largest contributor, dominating overall counts.

Functional defects – second most frequent, together with structural make up ~80% of issues.

Cosmetic defects – least frequent, but still relevant for customer satisfaction.

Structural & Functional Defects:

Prioritize root cause fixes for durability and functionality issues.

Implement stricter quality checks on these categories.

Most Problematic Machine/Shift:

Machines: A small set of product IDs (top 5) account for the majority of defects, with some linked to higher critical severity.

Shifts: Friday–Sunday shifts consistently show higher defect counts, indicating end-of-week underperformance.

High-Risk Machines:

Conduct preventive maintenance and operator retraining on top defect-prone machines.

Monitor defect severity trends per machine.

Trend Direction:

Monthly: Fluctuating, with January peak and alternating rises/drops afterward.

Quarterly: Q1 defect counts higher than Q2, suggesting early-year quality challenges.

Weekly: Spikes around weeks 17–20, likely tied to production cycles or campaigns.

Shift Management:

Investigate workload distribution and staffing on Friday–Sunday shifts.

Introduce mid-shift audits or process checks to reduce end-of-week spikes.

Financial / Operational Impact

High defect concentration in structural + functional categories implies **increased repair/rework costs** and potential **downtime.**

End-of-week shift issues may lead to **lower throughput** and **higher scrap rates**.

Targeting top machines and high-risk shifts could reduce defects by **50–80%**, yielding significant cost savings and efficiency gains**.**

Seasonal & Weekly Patterns:

Align production planning with known high-risk weeks/months.

Introduce targeted inspections during Q1 and weeks 17–20.